



Chemometric analysis for river water quality assessment at the intake of drinking water treatment plants☆

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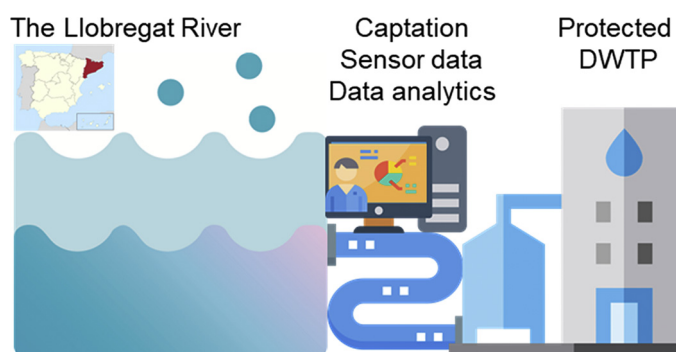
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HIGHLIGHTS

- New multiparametric approach for river water source quality monitoring at DWTP
- PCA modelling distinguished three factors to dominate the river quality.
- The organic and inorganic matter factors show time-series patterns.
- Events with organic and inorganic matter origins have been discussed.
- Microbiological and organic matter events occur simultaneously.

GRAPHICAL ABSTRACT



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ABSTRACT

Most of the Mediterranean rivers are suffering the effects of industrial, urban and mining discharges, as well as a reduction in water quantity and quality. Additionally, due to the Mediterranean climate, the natural water resource availability is periodically lower than the water demand in the area. Operation of drinking water plants in these geographical areas needs advanced process control systems where real-time and in-line water quality monitoring tools are key components. Data sets with parameters generated by monitoring sensors and from laboratory analysis are used to reveal the main factors that characterize water quality. Chemometric tools like Principal Component Analysis (PCA) can be used to explore and analyze correlations among different physico-chemical and microbiological parameters with the aim to assess the river water quality at the water intake of drinking water treatment plants (DWTPs). Strong seasonal trends in the organic and inorganic matter contents and unusual events in the raw river water quality at the DWTP water intake are revealed. Organic and inorganic patterns are then associated with climatological, meteorological and industrial pollution circumstances typical for the geographical region under study. In addition, microbiological events can be detected at the water intake of DWTP which may occur simultaneously with increasing water contents of organic matter, especially at the beginning of rainfall episodes. The application of PCA on sensors data in the water intake at DWTPs offers new possibilities for improved quality assurance and control procedures for DWTP management and its strategy.

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1. Introduction

Water resources management is an important element of sustainability in urban areas, with complex infrastructure networks. Moreover, this is even more relevant in Mediterranean semiarid regions with water scarcity scenarios in which water resources are affected by high salinity and the presence of micropollutants (Köck-Schulmeyer et al., 2011; Kuzmanovic et al., 2015; Momblanch et al., 2015). Advanced urban water infrastructures specially drinking water plants for such scenarios require a significant increase in capital and operational expenses (Aydin et al., 2015). The need for smart infrastructure represents a significant capital investment where highly efficient treatment technologies and in-line and real-time water quality monitoring tools are key elements (Mustonen et al., 2008; Marlow et al., 2013; Behmel et al., 2016). Efficient waterworks management requires clear goals and definitions, time and space scales, and quantifiable indicators.

Many urban areas in different regions and countries (e.g. California and Arizona in USA, Israel, Australia, Mediterranean coast in south Europe and north of Africa, Southern cities of South Africa) with drinking water supply networks based on surface water resources, have experienced a reduction of their water quantity and quality due to climate, environmental and anthropogenic pressures (McDonald et al., 2011; Rockström and Karlberg, 2010; Navarro-Ortega et al., 2015). Different infrastructures have been proposed and implemented to improve drinking water quality and even more important to ensure the long-term resources availability. The general approach is following the integration of processes where the role of membrane technologies is fundamental (Gibert et al., 2017; Raich-Montiu et al., 2014; Vera et al., 2017). Among these technologies the more important are membrane-based technologies in drinking water treatment plants (DWTPs) (e.g. nanofiltration, reverse osmosis, electrodialysis reversal) to diminish the salinity levels and to reduce micropollutants and seawater desalination technologies when surface and groundwater resources are scarce (Ferrer et al., 2016).

One well known example of this scenario is the Sant Joan Despí DWTP on the Llobregat River at the Barcelona (Catalonia, Spain) Metropolitan Area (BMA). The Llobregat River is one of the major resources of water for the Barcelona metropolitan area with 4.7 million of inhabitants. It supplies around 40% of the total water demand (Devesa et al., 2007).

Fig. 1A shows the location of the SJD DWTPs just 15 km west of Barcelona. The plant has >50 years of experience in producing and supplying drinking water in Barcelona. It comprises traditional plant operation schemes together with sophisticated membrane treatment technologies based on ultrafiltration and reverse osmosis (López-Roldán et al., 2016) converting the SJD DWTP into one of the most sophisticated DWTPs across Europe with water production capacity of 5.5 m³/s (see Fig. 1B).

Along the Llobregat River basin catchment area, there are three dams located in the upper part and several small rivers contribute to the main streamflow. In the upper and middle part of the river (Marcé et al., 2012), the mineral composition of water is strongly affected by sedimentary rocks such as limestone, marls, gypsum and halite, resulting in very high levels of mineralization (dry residue around 900 mg/L, Fernandez-Turiel et al., 2003). The heavy industrialization and agricultural growth along the river basin, the financial and technological constraints, and non-enforcement/bypassing of public laws, have led to considerable water pollution of the Llobregat river (Ginebreda et al., 2012). Frequently, the river water quality of this river is affected by accidental sewage discharges, industrial spills, and pollution from agricultural and urban run-off (Aguillera et al., 2012). Moreover, the Llobregat river water quality is strongly affected from the potash mining industry located in the middle part of the river basin (Ladrera et al., 2017) during raining episodes. Furthermore, the climate changing conditions in the region (Versini et al., 2016) produce difficulties in DWTP management. More specifically, scarce snow/

rainfalls along the year cause long and severe droughts, whereas a few heavy spring/summer rainfalls lead to floods with an adverse effect on water quality.

The main objective of this study is to demonstrate the possibilities of using chemometrics methods like PCA for the simultaneous analysis of multiple water quality parameters with the goal of adopting more comprehensive monitoring and water quality control strategies, reducing both time and efforts expend in the monitoring process. In the near future detection and analysis of events/patterns can be strongly facilitated by collecting data from multiple connected sensors for water quality assessment incorporating internet resources in DWTP management together with the methods proposed in this work for multivariate data analysis tools.

2. Material and methods

2.1. Current water quality monitoring practices at the intake

The existing practice for water quality monitoring at the intake of SJD DWTP is based on the inspection of individual raw water quality physicochemical parameters such as the UV absorption at 254 nm (UV₂₅₄), turbidity (TURB), total organic carbon (TOC), pH, conductivity (COND) and others. DWTP operators based their decisions on individual parameter values using pre-established thresholds. DWTP management is certified according to the ISO 22000 norm, ensuring continuous compliance of the final treated water. For the quality monitoring process, data obtained by grab sampling and from laboratory analysis and/or from on-line/in-line sensors are used. Decision-making process is largely univariate in nature, where statistical analysis is conducted by monitoring one by one the different physicochemical parameters at the water intake of DWTP. Many of these individual thresholds have been suggested by specific legislative directives to be fulfilled. Because such thresholds do not capture local climate conditions and site-specific events, the monitoring activities are subject to severe limitations. More specifically, the prevalent policy of using individual thresholds falls to consider that water quality is determined by the interaction of multiple parameters in water system. As a result, hidden and/or latent information about the correlations and interrelationships among these multiple parameters is not incorporated in the monitoring process. For instance, an event related to flooding can be described by the combination of different parameters such as turbidity, organic load and higher UV254nm absorbance. During rain events, the values of these parameters will increase significantly (Volk et al., 2002). A combination of increased values of conductivity, ammonium concentration, spectral abnormalities and unusual pH changes on the other side would suggest a chemical pollution event.

Water quality monitoring practices also include the microbiological control of the raw water, which is permanent at the water intake of DWTP. There are legislative standards for microbiological control of several bacteria species, which require their permanent monitoring (DWD, 1998). In SJD DWTP, microbiological analyses at each step of the treatment are performed once per week at minimum. At the end of the required time of proliferation (usually 1–2 days), bacterial colonies are identified and quantified. Presence of bacteria at the entrance of DWTP is chronic. The existing analytical procedure however does not account for sudden spikes of the bacterial counts with an accidental character. The knowledge about water quality parameters with positive correlations with bacterial events would help to design strategies for future monitoring and protection of the water intake.

Several contamination episodes were recorded at the SJD over the last past years. They were caused primarily by industrial discharges of dicyclopentadiene and derivatives (Ventura et al., 1997), dioxanes/dioxolanes (Romero et al., 1998) and creosote (Ventura et al., 1998). Boleda et al. (2007) reviewed taste and odor events detected from 1990 until 2004. These previous studies have concluded that the industrial contamination has produced more contamination episodes than

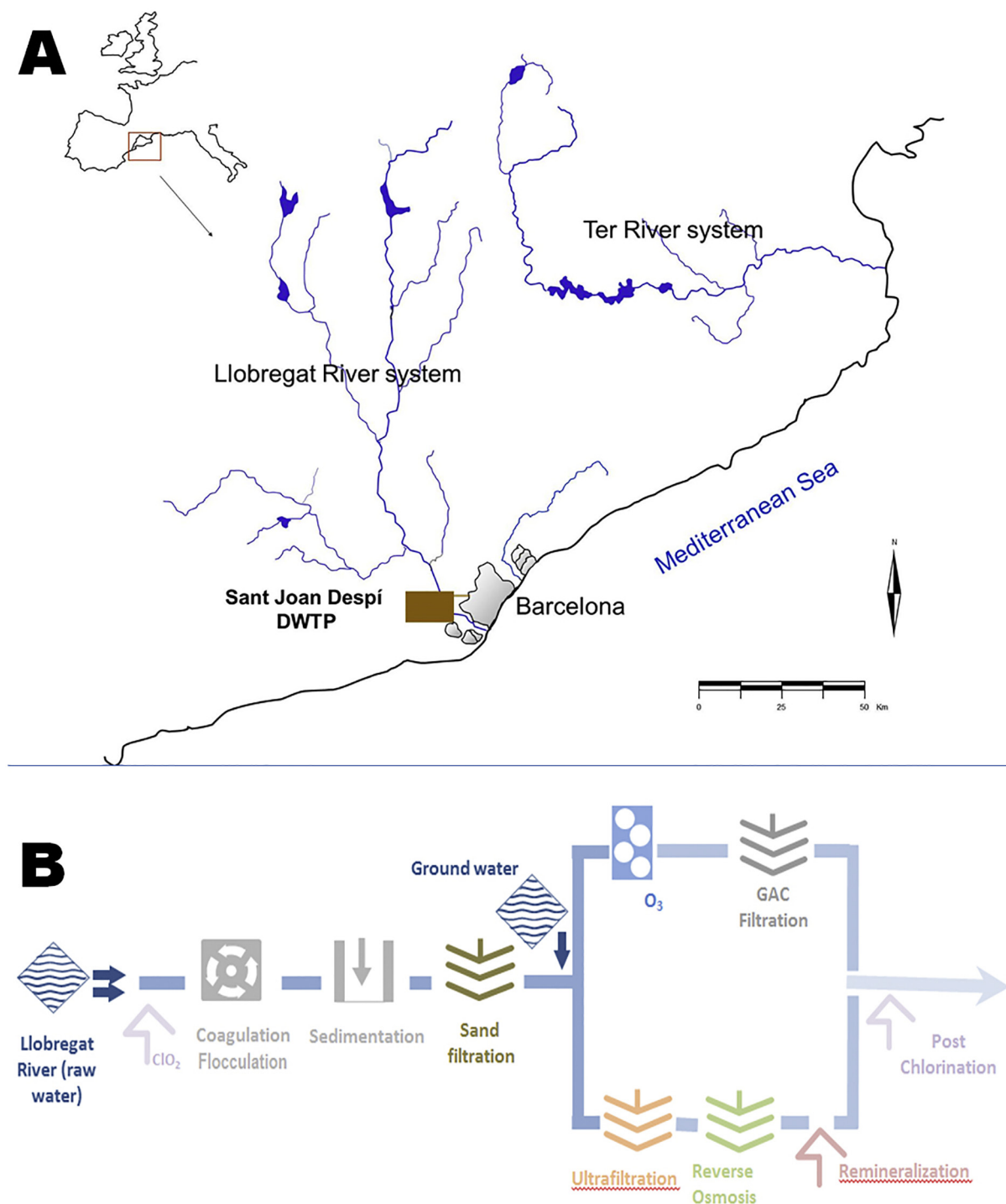


Fig. 1. A) The Llobregat river and the location of the Sant Joan Despí drinking water treatment plant nearby Barcelona, Spain. B) SJD DWTP scheme of the water treatment processes.

other factors. [Martin-Alonso et al. \(2007\)](#) reported a contamination episode with diacetyl and butanedione. The permanent quality monitoring at the intake of the Sant Joan Despí (SJD) drinking water treatment plant (DWTP) over the past years has generated a significant volume of data about the Llobregat River raw water quality. The analysis of these data sets can be performed using multivariate statistical methods. The information extracted from these analysis can be used to significantly improve water plant management and decision-making strategies and allow adoption of improved approaches for protect DWTP protection from river water pollution events.

2.2. Data organization

Historical data for 8 physicochemical quality parameters, monitored by on-line sensors in the water intake from the Llobregat River during two years 2013–2014 and stored in the Laboratory Information Management System of DWTP, were collected. The monitored parameters were turbidity in UNF (TURB), ammonium ion in mg/L (NH_4), total organic carbon in mg/L (TOC), UV absorbance at 254 nm abs/100 cm (UV_{254}), conductivity in $\mu\text{S}/\text{cm}$ (COND), pH and river flow in m^3/s . Additionally, microbiological parameters from laboratory assays were also considered.

Two aspects have been considered in the preparation of the different data sets analysed in this work. First, the availability of parameters measured in the plant and of their scheduled monitoring programs, in agreement with new Drinking Water Regulations, the DWTP, and the ISO22000, which establish the need of increasing the number of monitoring procedures. Second, the possibility of direct alignment of all parameters according to their recording time, avoiding losing of information during this alignment and the incorporation of missing values.

Collected data were arranged in three different data sets called respectively:

- a) *Dataset 1*: Time interpolated physicochemical observations,
- b) *Dataset 2*: Experimental physicochemical observations, and
- c) *Dataset 3*: Augmented datasets including microbiological parameters.

Although during year 2013 physicochemical parameters were measured daily, unfortunately they did not match in their recording time. The resulting data table had a considerable number of systematic missing. To obtain a data table without missing values, an interpolation algorithm is proposed to estimate them. Thus, the application of such as algorithm resulted in Data set 1 with the possible inconvenience of losing some information about the parameters real interactions due to the artificial data filling step. By the mid 2014, all required physicochemical sensors at the intake were aligned according to the time of their recording, and therefore, the real interactions among parameters could be investigated directly in Data Set 2. The third data set was acquired according to the microbiological sampling schedule in DWTP, e.g. every Monday at 08:00 AM. This data set allowed to find out the correlation between the microbiological results and with the corresponding physicochemical parameters.

In *Dataset 1*, a linear interpolation was preliminary used to align the parameters on an hourly basis. The final data table included 16,333 hourly measurements for 7 parameters. This dataset has allowed the evaluation of the temporal variations in water quality over the investigated period.

In *Dataset 2*, only those physicochemical parameters with real experimental observations were considered and used for analysis. This dataset has observations of the 7 parameters recorded at exactly the same time during the period from July to November 2014. In total, it includes 346 experimental observations (several measurements per a single day), matched in time for the 7 parameters. This data set was used for the evaluation and validation of the changes in the water quality trends, events and relationships between parameters initially detected from the comprehensive analysis of *Dataset 1*.

In *Dataset 3*, microbiological data were collected on a weekly basis for 2014 (52 samples). The microbiological information of this dataset includes bacterial counts of *Pseudomonas* sp., *Enterococcus* sp., *Escheria coli* sp., Total coliforms, *Clostridium* sp. and *Aeromonas* sp. This dataset has been rearranged to match the corresponding physicochemical data recordings. In this new dataset, the information about the physicochemical parameters and river flow were also considered in the analysis. Thus, this new augmented data matrix considered a total number of 52 samples (one per week) and 14 parameters (6 microbiological indices and 8 physicochemical parameters).

2.3. Multivariate statistical analysis of water quality parameters at the intake of the SJD DWTP using principal component analysis

Multiple parameters, generated by on-line sensors and from laboratory assays generate large sets of samples, which can be simultaneously analysed to provide a comprehensive view of natural raw river quality patterns and unknown events and hence provide adequate tools for the improvement of water treatment procedures in DWTPs. Datasets

are stored in two-dimensional tables or data matrices, where observations/measurements (ordered by time or by samples) are in the rows and measured parameters/variables are in the columns of a data table (data matrix). Thus, these two-dimensional data tables can be easily analysed using existing multivariate statistical and chemometrics methods.

Principal Component Analysis, PCA, (Jolliffe, 2002) is probably the most popular method in multivariate statistical analysis of environmental data. It is based on the assumption that in the original data set there is a small number of dominant factors with significant influence (components) which are the main source of variation. These factors cannot be measured directly, because they cannot be captured with a sensor or observed experimentally. As a result, they are usually called hidden factors. Thus, PCA would allow the investigation of temporal (seasonal), meteorological and pollution patterns/trends in the different recorded data sets. PCA allows for the investigation and modelling of the complex relationships and/or interactions existing among multiple parameters and samples (observations) in the analysis of environmental monitoring data sets. PCA results can show the hidden underlying processes in the monitoring of water quality parameters in DWTPs.

According to the PCA model, the original experimental data matrix **D**, is decomposed using a bilinear model, giving two orthogonal matrices, **T** scores (mapping the samples on the principal components), **P^T** loadings (mapping the measured variables on the principal components) and **E** is the matrix of residuals (unexplained variance) as in Eq. (1).

$$\mathbf{D} = \mathbf{TP}^T + \mathbf{E} \quad (1)$$

The number of principal components to be considered in the PCA model (rows of **T** and columns of **P^T**) throughout this study was selected on the basis of the following criteria: a) the sizes of the eigenvalues associated with the principal components, and b) a meaningful environmental explanation of score and loading profiles. In case of water quality data, **P^T loadings** display the physicochemical profiles of the principal components and **T** scores give the mapping of the sample projections on them. In this water quality study, the scores will give important information about the temporal distribution (measurements over time, time distribution) of the samples, which I show possible hidden trends and can be analysed by time series analysis (see below).

Unusual events at the water intake of the San Joan Despi DWTP are investigated. An effective and useful way for the detection of these unusual events is the Q residuals chart, which shows the differences (or residuals) between measurements and their projections on the k principal components retained in the PCA model along time. Q residuals indicate the degree to which a particular measurement conforms to the model. Samples with large residuals values are not well explained by the model and they can be considered as unusual events. When a measurement is flagged as an event, it is recommended to examine the so-called Q contribution plot (Wise and Gallagher, 1996). Q contributions plot displays one row of the residuals **E** matrix for the considered sample event. Q contributions plots show how much of every variable/parameter contributes to the overall Q value for the considered sample, while retaining the sign of the contribution of the given parameter. Visualization of parameter Q contributions can be used to identify which of them contribute more to the given sample's sum-squared residual error (Wise et al., 1989). Confidence limits can be also calculated for further examination. They provide threshold values of the regular/outlying conditions. When the quality of the water in the intake is under control, Q residuals should display small values within these confidence limits. Unusual (outlying) events are displayed outside these confidence limits

2.4. Curve fitting of times series

PCA of *Dataset 1* showed a periodic form, representing the score values at different time points t_i (on hourly basis), which is assumed to

be a mixture of the useful temporal signal y (the general trend in the water quality over the entire period of monitoring for each of the principal components) and noise, which would be the water quality non-time dependent periodic events or extreme values (Deng and Wang, 2017). The useful signal y can be represented as a continuous periodic function defined over a certain time interval and the water quality values not explained by this periodic function are considered to be noise. This periodic nature of the scores (as periodic signal $g(x)$) can be modelled by a Fourier series approximation (Mueller-Warrant et al., 2012) according to the equation:

$$g(x) = a_0 + \sum_{i=1}^n (a_i \cdot \cos(i \cdot w \cdot x) + b_i \cdot \sin(i \cdot w \cdot x)) \quad (2)$$

where a_0 models a constant (intercept) term in the data and is associated with the $i = 0$ cosine term, w is the fundamental frequency of the signal, n is the number of terms (harmonics) in the series, and $1 \leq n \leq 8$.

The best data fitting were obtained by visual inspection and considering the highest determination coefficient as measure of how well the above mentioned equation performed as a predictor.

3. Results and discussion

The objective of this study is the modelling of the water quality parameters at the intake of the SJD DWTP (Dataset1, Dataset2 and Dataset3) using Principal Component Analysis (PCA).

3.1. Seasonal variability of the Llobregat river water quality at the intake (PCA of dataset 1)

PCA was applied first to the autoscaled Dataset 1, containing the 16,333 hourly measurements of the 7 quality parameters (after time interpolation alignment). Results with 3 principal components (PCs) are shown in Fig. 2, explaining 72% of the total experimental data variance (PC1 35%; PC2 24% and PC3 13%).

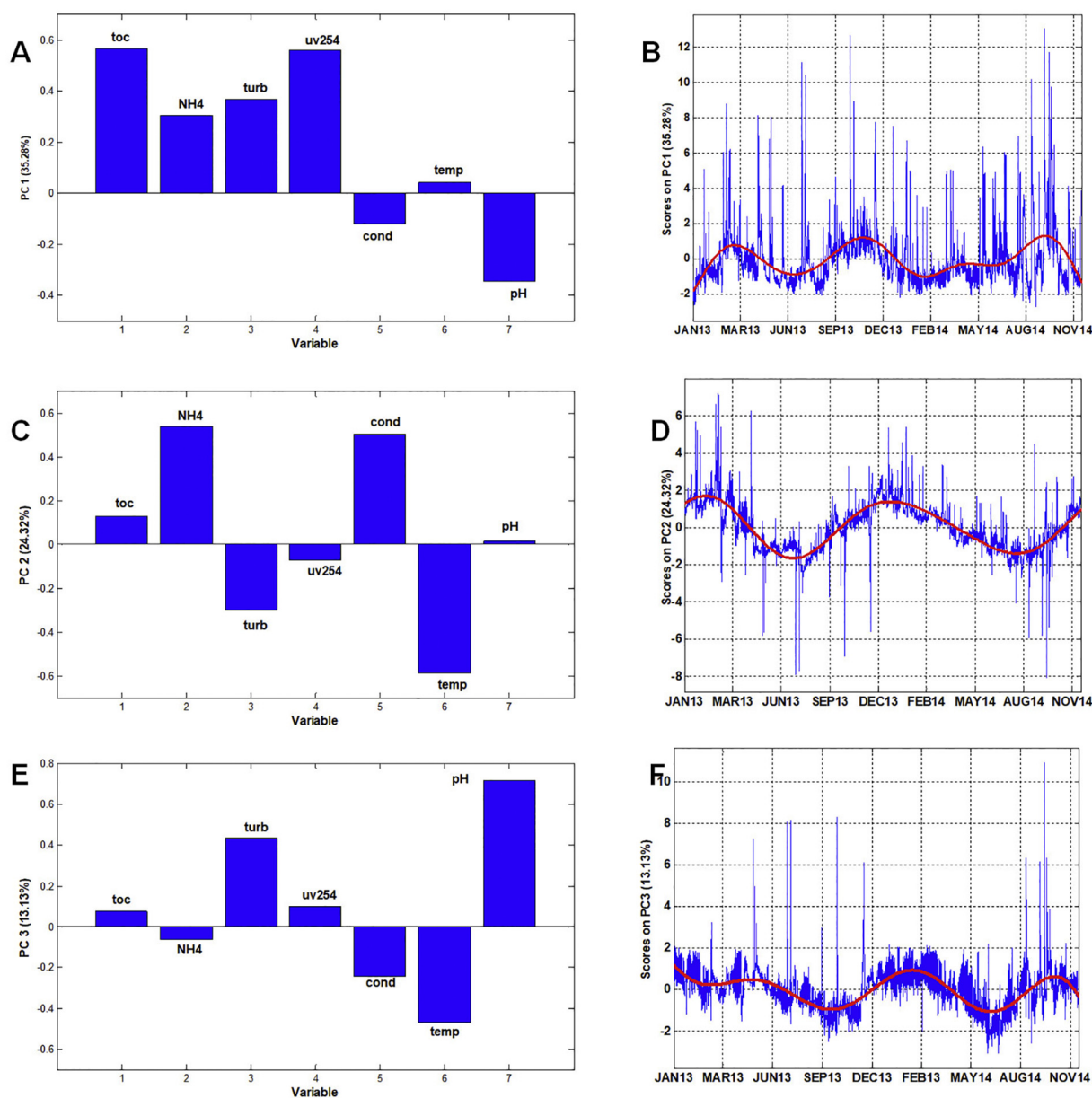


Fig. 2. PCA of the sensors data set for 7 physicochemical parameters at the intake of SJD DWTP obtained in two years of monitoring and applied linear interpolation for time alignment within sensor measurements. PC1-3 loading plots: A), C) and E); PC1-3 sample (hourly recordings) scores plots B), D) and F).

PC1 (35%) describes the variability of the organic matter as the main factor in Dataset 1 since it is strongly dominated by the positive correlation between TOC and UV_{254} (with the highest positive loadings on PC1). In addition, this PC1 reveals positive correlation of NH_4 and TURB parameters (also found with positive loadings) with the organic matter. pH had a negative loading and it was found to be inversely correlated with the other four above mentioned parameters (see Fig. 2A). The organic matter demonstrates a seasonal, temporal variability of the water quality (see Fig. 2B, PC1 scores on the right) and also sharp fluctuations occurring over the entire period of investigations. Positive PC1 scores were observed for measurements in spring and in autumn, whereas scores with negative sign were found in summer and winter. Also, positive scores were found for the sharp short termed positive fluctuations on this PC1. This observation confirms the expectation that the concentrations of organic matter were above average in spring and autumn, and during the sharp fluctuations, while the concentrations of organic matter were below average for summer and winter.

PC1 scores were fitted to the Fourier model, described in the method Section 2.3, with a R^2 value of 0.97. The sigmoidal trend-line in the organic matter content for 2013 showed two peaks. This fact can be explained by the typical Mediterranean type hydrological regime (nivo-pluvial) of the Llobregat River, which also has two maxima of the river flow- in spring and in autumn as shown in Fig. 3A, where the 2013 monthly accumulated precipitation is plotted. A considerable reduction of the organic matter in the Llobregat River is observed in the middle of summer and winter seasons in 2013. This phenomenon was weaker in 2014, largely due to anomalies related to a persistent drought observed for the spring of 2014 as seen in Fig. 3B and affecting the Llobregat river hydrological regime during this year. The short term fluctuations observed with positive scores in Fig. 2B, can be explained by the sudden sharp increases of the organic matter content (TOC and UV_{254}), leading also to the increase of TURB and NH_4 parameters when large flows of water arrive to the plant due to rain episodes in the river basin. The inverse correlation between them and pH can be explained due to the dilution effect produced by the high river water flows during spring, autumn and rain episodes.

PC2 scores and loadings outlined another type of temporal variability in the water quality trend. The second most important factor to consider in the evaluation of the the water quality changes is related with the variability of the inorganic matter content. On the loadings plot of Fig. 2C, ammonium ion and conductivity have the high positive loadings, whereas the rest of parameters have negative loadings. In Fig. 2D, a temporal sigmoidal trend line is also observed (in red, the fitted Fourier model gave in this case a R^2 value of 0.91), suggesting also a time-series behaviour of this factor. Positive scores on PC2 were

observed for the measurements in winter, whereas the scores for observations in summer were observed with negative sign values. In winter season, the river water quality was characterized with higher conductivity and higher ammonium ion concentrations in comparison to the summer season. During winter seasons, the Llobregat river is characterized with low water flows resulting in higher concentrations of inorganic matter. On the contrary, in spring and in autumn, when water flows increases, the concentrations of inorganic matter were around their average values. Short-term sharp fluctuations were also observed along the trend line of PC2 scores. Such fluctuations in the water quality in the intake of the water treatment plant were associated with the events of unusual changes in the inorganic content (Fig. 2D), when very high values of the measured conductivity and of ammonium concentrations were identified.

PC3 (13% of explained variance) explains the pH variability of the river water (strong positive loadings on PC3, Fig. 2E). In this case, it is difficult to attribute a seasonal trend related to seasonal changes, although there is some small temporal changes that could be fitted with the Fourier model with a R^2 value of 0.90 for this period of monitoring (red line in Fig. 2F).

On this PC 3 turbidity shows positive correlation with pH. This fact revealed that there were fractions of suspended material at basic pH values of the river water. On the contrary, on PC 1 (35% of variance explained) turbidity was found inversely correlated with pH. This is due to the increased contents of organic matter and suspended solids during the spring and autumn due to the rainfalls, and the rise of the river water flows lead to a dilution effect and decreased pH values.

3.2. Event detection in the intake of the Llobregat River (PCA of dataset 2)

The focus of this particular study is on the short-term, strong fluctuations, which can be assigned to possible pollution events, detected in the PCA scores plots.

Dataset 2 with the 346 real experimental observations for the seven parameters (which matched their time recordings) was analysed with PCA to further investigate and validate the nature of the previously observed events (in Dataset 1) related with the organic matter, inorganic matter and pH component sources explained in the previous section. These 346 observations spanned six months of monitoring, between July and November 2014. PCA applied to the autoscaled data suggested a model with three principal components, explaining as much as 81% of the total data variance. The relative percentage variance contributions of these components were: PC1 40% > PC2 25% > PC3 16%.

Analysis of PC1-PC3 loading plots (see Fig. 4A, C and E) confirmed that the same three factors previously explained in the analysis of

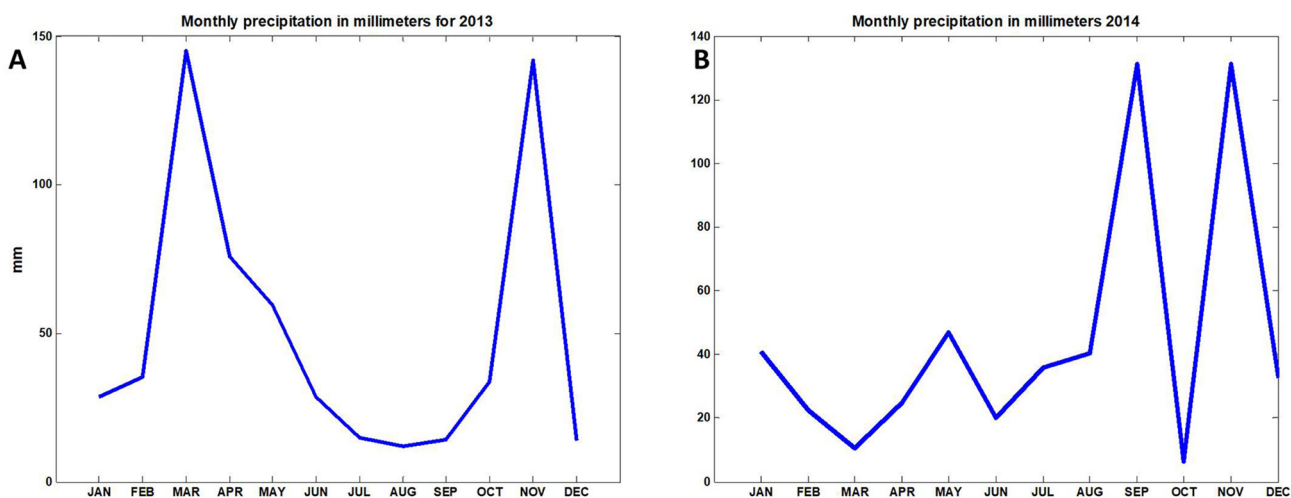


Fig. 3. Accumulated monthly precipitations registered for 2013 (A) and for 2014 (B) at the automatic meteorological station of El Prat de Llobregat, 5 km from the Sant Joan Despí DWTP.

Dataset1 are present in Dataset 2. PC1 describes the variance ion the organic matter content and detects the more significant events related with sudden change in its concentration. PC2 describes the variance in the inorganic matter and also the events related with its sudden change. PC3 is related with the river water pH changes. A deeper discussion of these three factors is given in Section 3.1. A similar behaviour is confirmed for the two data sets (Dataset 1 with interpolated data and Data set 2 with real data over the same time period). Whereas Dataset 1 was especially useful for the exploration of the general water quality trends Dataset 2 was useful for the investigation and comparison of the short term events with significant impact on the water quality.

The following discussion is focused on three consecutive events observed on the scores plots of Fig. 4 and labelled with numbers 1, 2 and 3.

The selected three events appeared at the same time on each of the three PCA scores plots suggesting that these three principal components contributed independently to each of the mentioned events. These events also show very high Q residuals (see Fig. 5A) suggesting extreme values. Further analysis of the Q parameter contributions plots (Fig. 5B, C and D) allowed to correlate these three events to a particular natural (meteorological) or industrial (spills) phenomena using meteorological information from weather data bases website meteoclimatic.com. Weather information from numerous meteorological stations around

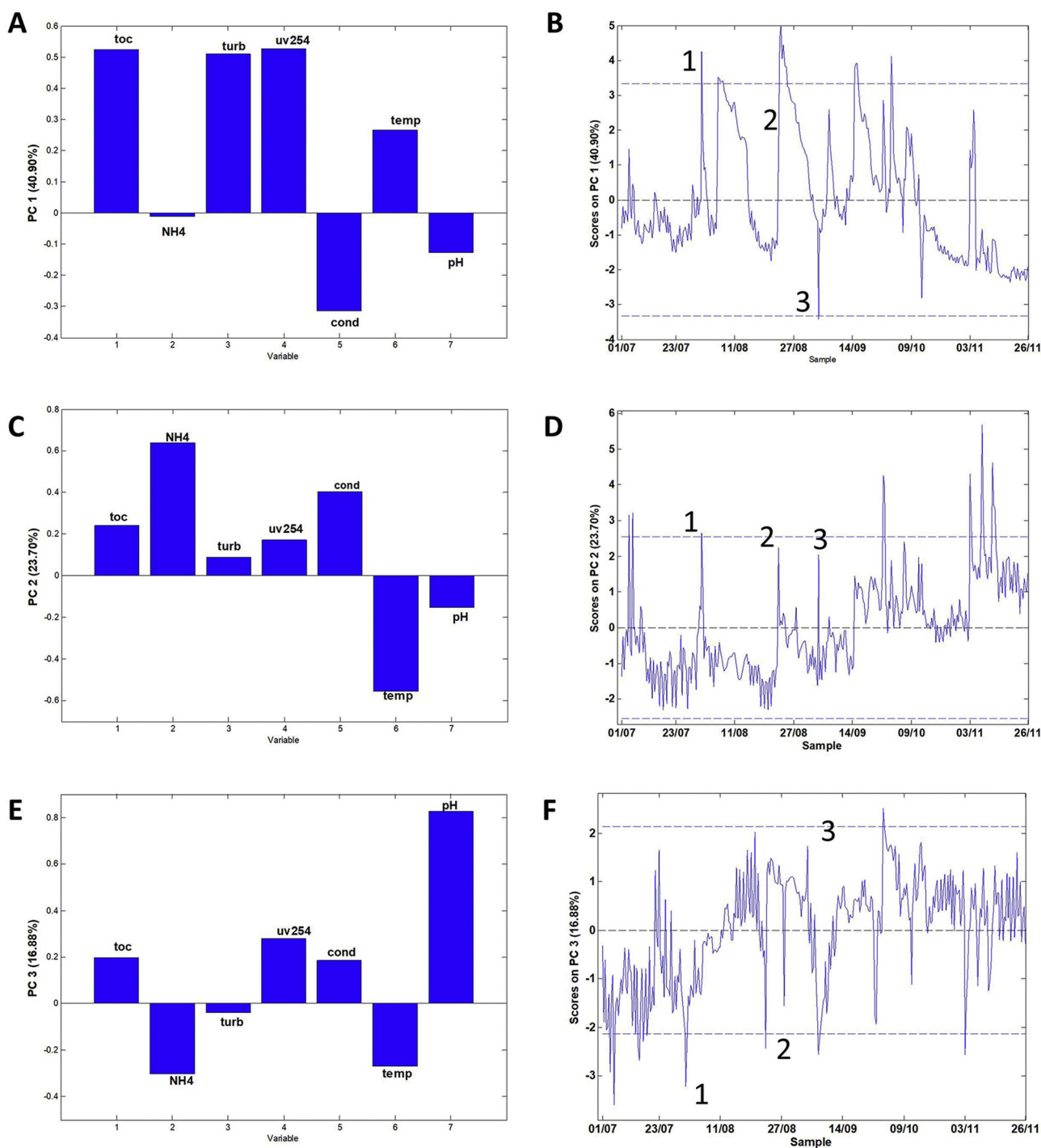


Fig. 4. PCA results of dataset containing 346 observations corresponding in their time of measurement for the seven parameters. PC1-3 loading plots: A), C) and E) respectively, and PC1-3 sample scores plots: B), D) and F) respectively; Three events have been pointed out with numbers on the scores plots.

Catalonia is provided systematically on this website. Data from stations spread over the whole region are stored chronologically and are publicly available. Thus, it is possible to examine for correlations between our event detection with the local weather information from this website.

Event 1 occurred at the river water intake on 03/08/2014 at 06:00 h. The sensor readings at this hour recorded TOC 7.1 mg/L; NH_4 1.91 mg/L; TURB 567 UNF; UV_{254} 15.8 abs/100; COND 784 $\mu\text{S}/\text{cm}$; TEMP 22.9C and pH 7.6. For this Event 1, the PCA model reported unusual sample Q-residuals with extreme values (unexplained variance) for turbidity as seen in Fig. 5B. On 01/08 and 02/08/2014, the information collected at the weather webpage identified several rainfall episodes occurred in 24 h with recordings up to 30 mm of rainfall (see Fig. 6). Thus, this event 1 was an example of the effect of a sudden increase of the washed out natural organic matter into the river flow. Simultaneously, large volumes of rain water led to strong dilution effects, also for the minerals, resulting in a significant reduction of COND and pH parameters respectively (found with negative loadings on PC1). The rest of the event peaks with strong positive scores on PC1 were also associated to rainfall events during the six month period of investigation. Events with high organic matter associated to industrial discharges on PC1 were not detected.

Event 2 defined the beginning of a new episode of heavy rain with a sharp rise in the concentration of organic matter detected on PC1. This organic matter event was gradually diminishing after some days on this plot giving the beginning of the next event 3, observed as strong event on PC2 and PC3 plots and explained further in the next section. Event 2 occurred at the DWTP intake on 22/08/2014 at local time 22:00 h. The sensor readings recorded were TOC 6 mg/L; NH_4 1.91 mg/L; TURB 180 UNF; UV_{254} 13.9 abs/100 cm; COND 1045 $\mu\text{S}/\text{cm}$; TEMP 23.2C; pH 7.8. The Q parameter contributions of the PCA model revealed unusually high values for NH_4 and in a less extent for pH and water temperature (Fig. 5C). This rainy episode event started with a sharp rise on NH_4 suggesting a possible runoff from the wastewater treatment plants located nearby. Event 2 on 22/08/2014 finished in the beginning of event 3.

The peak of the event 3, occurred at 03:00 h on 01/09/2014 with sensor recordings: TOC 2.6 mg/L; NH_4 0.24 mg/L; TURB 153.3 UNF; UV_{254} 11.2 abs/100 cm; COND 3413 $\mu\text{S}/\text{cm}$; TEMP 24.8C; pH 8.4. This event shows unusual Q parameter contribution (Fig. 5D) only for the COND parameter. In contrast, the rest of parameters were within their regular levels. This event was produced by the unusual increase of concentrations of dissolved minerals in water. Conductivity value was recorded above 3400 $\mu\text{S}/\text{cm}$ and, as a result, the DWTP had to close its activity of water intake. Valuable information about weather conditions during event 3, was retrieved from the meta-analysis of the meteorological database. During 30/08/2014 and 31/08/2014, multiple local heavy rainfalls (up to 25 mm of accumulated rainfall in short time) were recorded by the meteorological stations located in radius of 30 km near the potash mining facilities nearby the river basin. It is plausible therefore to suggest that these heavy rainfalls also affected the industrial area and produced significant mineral spills which were detected at the DWTP intake as an unusual event.

3.3. Correlating microbiological data with the physicochemical events in the water intake (Dataset 3)

Data from the investigated seven physicochemical parameters previously analysed in Dataset 2 were augmented with the river flow and microbiological data for the same 52 days giving Dataset 3. A total number of 14 variables were considered (6 bacteriological, River flow and the 7 physicochemical parameters). All the variables were autoscaled. The model with 3 PCs explained >80% of the total data variance and the first PC already accounted for >50% of the variance. This first PC captured most of the variation for all bacteria counts and for some of the physicochemical parameters such as river flow and TURB.

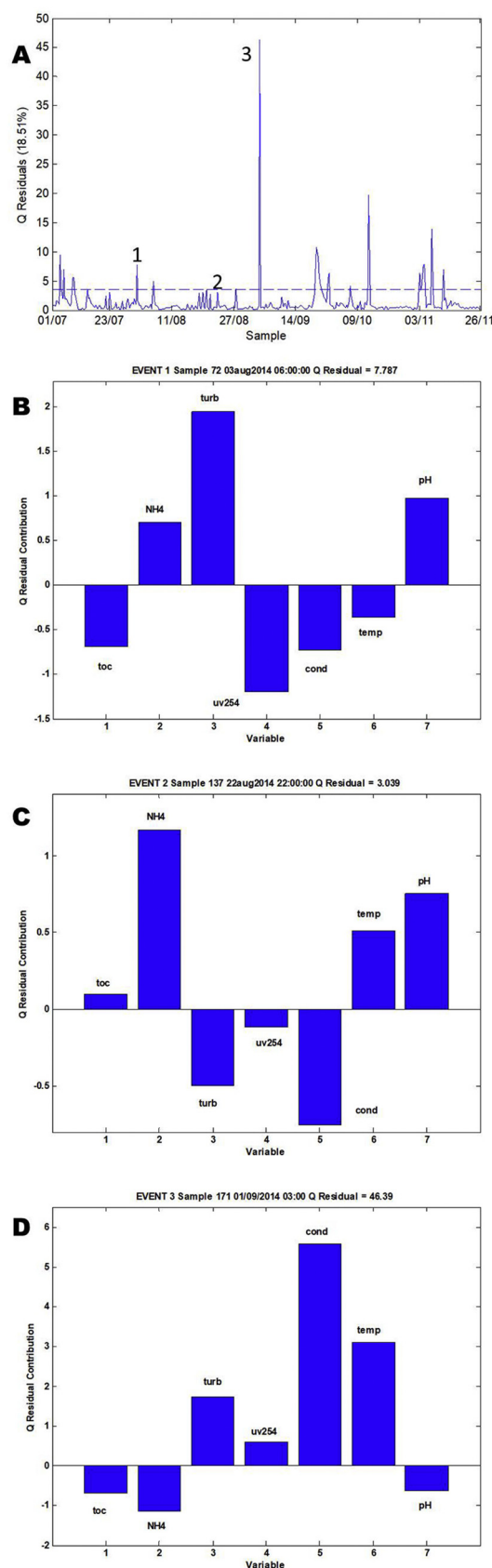


Fig. 5. Three events associated with high Q residuals for the PCA model with 3 components: A) Q residuals for PCA; B) Q residual contributions for parameters of event 1; C) Q residual contributions of parameters for event 2; D) Q residual contributions of parameters for event 3.

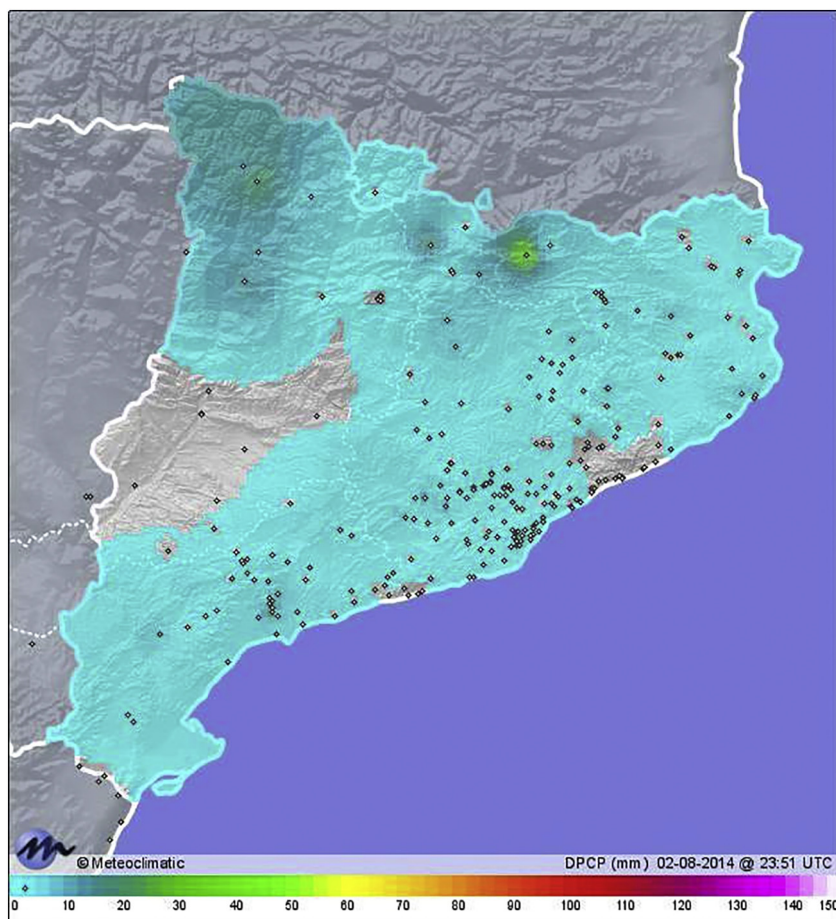


Fig. 6. Accumulated rainfall on 02/08/2014 for 24 h in Catalonia (Spain). Source: www.meteoclimatic.com

PC1 scores (in Fig. 7A) figured out different events occurring during a particular week in the water quality at the intake. The analysis of the loadings plot of Fig. 7B reveals that all 7 bacteria concentrations, together with the river flow, TURB and UV₂₅₄, have high positive loadings on PC1 meaning strong positive correlation among them. In contrast, inverse correlations between bacteria concentrations and COND and pH water parameters were observed. This result suggests that high bacteria concentrations were correlated with lower conductivities and pH values in water. Peak values in the concentrations of all monitored bacteria occurred when the river flow was high, with strong turbidity and elevated values of UV₂₅₄ absorbance. In these conditions, COND and pH values

were lower due to dilution effects—suggesting rainy episodes. Therefore, Most probably rainfalls washed out river biofilms, and produced agricultural and urban runoffs into the river water, contributing to a significant increase of the observed microbiological events. In addition, wastewater overflow from WWTPs collectors along the Llobregat river basin is feasible during these events.

Five events with strong bacterial occurrence were observed for the investigated period. The strongest one was detected on Monday - 47th week/sample. There were also other less intense events during 5th, 17th, 20th and 40th week of the monitoring period. Investigation of meteorological conditions (website www.meteoclimatic.com) for the

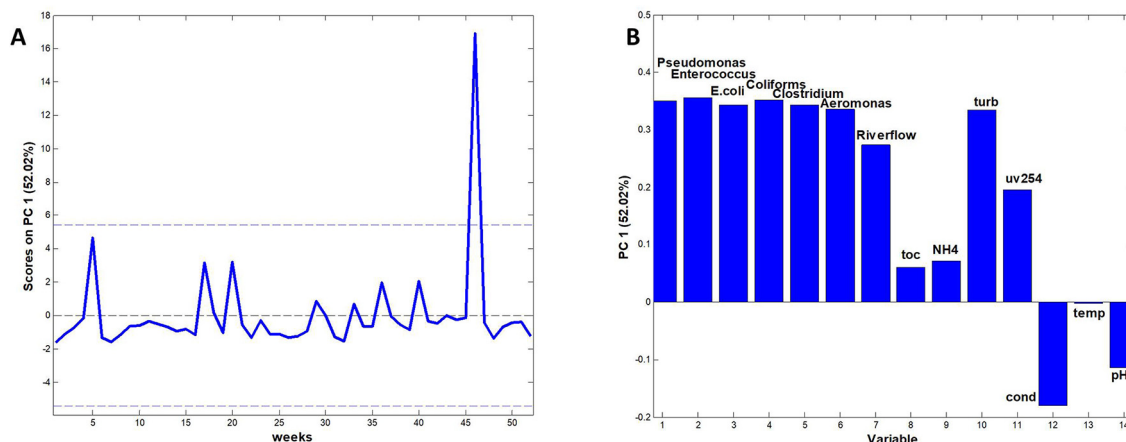


Fig. 7. PC1 results of dataset containing microbiological and physicochemical data. A) PC1 scores plot; B) PC1 loadings plot.

corresponding dates of detected events confirmed the existing correlation of the events of 17th, 20th, 40th and 47th weeks with rainfalls. The event occurring during the 5th week was not associated to particular weather anomalies, but in this case, a high ammonium concentration in the river water was suggesting a possible WWTP discharge of contaminated water.

3.4. Practical recommendations

Some practical recommendations can be considered. The concept of multivariate statistical process control (MSPC) with PCA has been shown to be a powerful approach (Ferrer, 2007) to monitor the continuous flow of information acquired by the physicochemical sensors in real time for the detection of undesired events, quality assurance, process optimization and process control. We propose to use PCA for real time control of the water intake processes in SJD DWTP. A new quality threshold can be calculated from the Q residuals of PCA model (Kourti and MacGregor, 1996). This new threshold will allow a better monitoring of the river water quality than the methodology used at present based on the monitoring of separated thresholds for all individual parameters implemented in the DWTP. This new threshold will take into account the observed correlations among the different physicochemical parameters. Thus, DWTP decision making will be focused not only in one parameter, but in all of them included in the PCA model, leading to a safer and faster response in case of undesired event detection.

The observed seasonal variability in the water quality of the Llobregat river can serve as an illustrative example for the design and planning of the proper adjustments of the operational procedures of the DWTP, as for example those related with the predominance of organic matter content in spring and autumn seasons, or those related with the higher inorganic content in summer.

4. Conclusions

Principal Component Analysis (PCA) of river quality monitoring data is shown to be an efficient chemometric tool to explore the complex relationships occurring between physicochemical and microbiological river water quality parameters, either monitored with sensors or obtained by detailed analysis in the laboratories of DWTPs. Organic matter events were characterized as a combination of three parameters with increased values: TOC, UV₂₅₄ and turbidity, whereas the events with inorganic character showed typical increases of ammonia, conductivity and pH. Pollution events were also correlated with recorded meteorological data. Organic and inorganic matter contents are the main factors that dominate the changes in the river water quality and they are mainly associated with climatic and meteorological circumstances. For example, organic matter contribution usually shows seasonal dynamics due to the nivo-pluvial hydrological regime of the Llobregat river. Its contribution is higher at the beginning of episodes of heavy rains. In contrast, the inorganic matter content shows an annual cycling trend with maximum values in winter, whereas other punctual events in the water intake with high inorganic matter content can occur also at the beginning of rainy episodes (as in the rise of NH₄ concentrations), or at the end of these events with a considerable increase of the conductivity. All these results suggest that multivariate statistical process control strategy using PCA can be used as a new strategy for the improvement of DWTP management and of the existing monitoring processes. The application of the proposed monitoring strategy of measuring and analysing simultaneously different parameters is of great practical help in the study and understanding of the effects of larger time scale climatic anomalies such as droughts. It can also be used for the monitoring and control of irregular events, at short time scale, such as those occurring from flush rains and industrial spills. This would allow faster operational responses to different type of pollution events and allow a better protection of water plant installations. Operational management efforts should be extended to monitoring of

temporal (seasonal) water quality variability and better control of the periodic pollution events from organic and inorganic sources. An improved control of these two sources would result in an easier control of microbiological events, as they appear especially during the beginning of rainy episodes. The combination of all investigated parameters can be used as specific fingerprints for the detection of river water pollution events.

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